Discrete Mathematics and Its Applications

course code: 211B0010

Spring & Summer, 2023

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Introduction

- **Course policies**
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  - **Textbook**
  - **Grading scheme**
- **Course overview**
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  - **Course goals**

Course policies

- **Instructor information:**
  - **Weng, Yanlin 翁 彦琳**
  - **Email: weng@cad.zju.edu.cn**
  - **Dingtalk group**
- **Textbook:**

  K. Rosen, *Discrete Mathematics and its Applications*, *Eighth Edition*, McGraw-Hill
- **Lectures:**
  - **Tuesdays 3-4 (10:00-11:35) & Fridays 3-4 (10:00-11:35)**
  - **Related courseware is available online after each lecture at the "Learning in ZJU" site (http://course.zju.edu.cn/) & the Dingtalk group**
- ***Web resources:***
  - **Companion website: www.mhhe.com/rosen**

Course policies

- **Grading Scheme**
  - **Homework (10%):**

    Two assignments per week and both are due at the beginning of the following Tuesday class

    NO LATE SUBMISSION ALLOWED UNLESS INFORMED IN ADVANCE
  - **Quizzes (20%)**
  - **Midterm exam (20%)**
  - **Final (50%)**

Course Overview

Discrete mathematics is an essential tool in almost all subareas of computer science.

- **Course goals**
  - Mathematical reasoning
  - Combinatorial analysis
  - Discrete structures
- **We will cover**
  - Fundamentals (chapters 1 – 5)
    - **Logic and Proofs**
    - **Basic structures**
  - Discrete Structures (Chapters 6 – 11, except 7)

    Counting (Chapters 6, 8), Relations (Chapter 9)

    Graphs (Chapter 10), Trees (Chapter 11)

Chapter 1
The Foundations: Logic and Proofs

1.1 Propositional Logic

1.2 Applications of Propositional Logic

1.3 Propositional Equivalences

1.4 Predicates and Quantifiers

1.5 Nested Quantifiers

1.6 Rules of Inference

1.7 Introduction to Proofs

1.8 Proof Methods and Strategy

Proposition

【**Definition**】**A** proposition is a declarative sentence that is either true or false, but not both.

- Examples
  - Where am I? x
  - Cheer up! x
  - X+1=2 x
  - 1+1=2 √
  - Hangzhou is the capital of China. √
  - The teacher exclaimed, “Don’t come into class late again!” √

Propositional Logic

- Propositional Logic (Calculus): the area of logic that deals with propositions
  - Propositional variables: small letters such as *p, q, r, s* …
  - Truth value: T (true proposition), F (false proposition)
  - Logical operators (Connectives): be used to form compound propositions from existing propositions
    - **Negation operator ¬ (NOT)**
    - **Conjunction operator ∧ (AND)**
    - **Disjunction operator ∨ (OR)**
    - **Exclusive or operator ⊕ (XOR)**
    - **Conditional operator → (IF--THEN)**
    - **Biconditional operator ↔ (IF AND ONLY IF)**

Negation (NOT)

【**Definition**】Let p be a proposition. The negation of p, $\neg p$, is the proposition "It is not the case that p."

- *Example:*
  - ***p*: Today is Tuesday.**
  - **$\neg p$: It is not the case that today is Tuesday. (Today is not Tuesday.)**

- Truth table

p	$\neg p$
T	F
F	T

The truth value of $\neg p$ is the opposite of the truth value of p

Conjunction (AND)

【**Definition**】Let p and q be propositions. The conjunction of p and q, $p \wedge q$, is the proposition "p and q."

- Example:
  - ***p*: Today is Tuesday.**
  - ***q*: It is raining.**
  - ***p* ∧ *q*: Today is Tuesday and it is raining.**
- Truth table

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Both *p* and *q* must be true for *p* ∧ *q* to be true

Disjunction (OR)

【**Definition**】Let p and q be propositions. The disjunction of p and q, $p \vee q$, is the proposition "p or q."

- Example:
  - ***p*: Today is Tuesday .**
  - ***q*: It is raining.**
  - ***p* ∨ *q*: Today is Tuesday or it is raining.**
- Truth table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Inclusive or: the disjunction is true when at least one proposition is true.

Exclusive OR (XOR)

【Definition】 Let p and q be propositions. The exclusive or of p and q, $p \oplus q$, is the proposition "p or q, but not both."

- Example:
  - ***p*: Today is Tuesday.**
  - ***q*: It is raining.**
  - ***p* ⊕ *q*: Either Today is Tuesday or it is raining, but not both.**
- Truth table

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Exactly one of *p* and *q* is true for the XOR to be true.

Example

- Consider the following propositions

p: Today is Tuesday.

q: I will go to the beach.

$p \wedge q$: Today is Tuesday and I will go to the beach

$p \vee q$: Today is Tuesday or I will go to the beach

$p \oplus q$: Either today is Tuesday or I will go to the beach, but not both.

$\neg p \wedge q$: Today is not Tuesday and I will go to the beach

$\neg p \wedge \neg q$: Today is not Tuesday and I will not go to the beach

Conditional Operator (If--then)

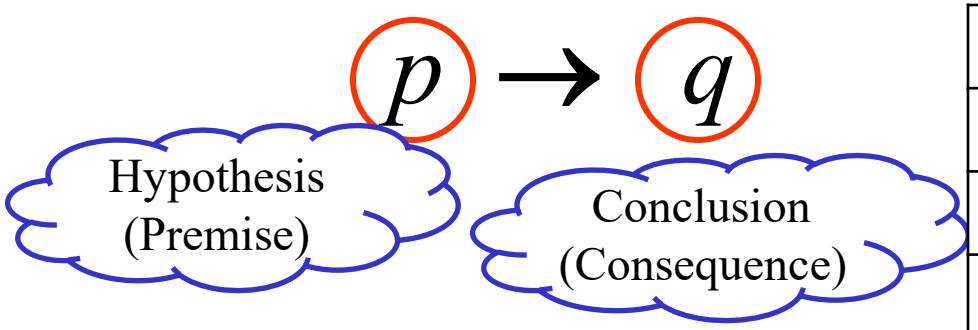
【**Definition**】Let p and q be propositions. The conditional statement (implication) $p \rightarrow q$, is the proposition "if p, then q. "

- Example:
  - ***p*: Today is Tuesday.**
  - ***q*: It is raining.**
  - **$p \rightarrow q$: If today is Tuesday, then it is raining.**
- Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The conditional statement is false only when *p* is true and *q* is false

Conditional Statement



p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- Examples:

If today is Tuesday, **then** 1+1=2. √

If today is Tuesday, **then** 1+1=3. X

If today is Monday, **then** 1+1=2. √

If today is Monday, **then** 1+1=3. √

- *If pigs fly, then you can get an A in Discrete Mathematics!*

Conditional Statement - Equivalent Forms

- If p, then q
- p implies q
- If p, q
- q if p
- q when p
- q follows from p
- q whenever p
- p is a sufficient condition for q
- q is a necessary condition for p
- **p only if q**
- q unless $\neg p$

Converse of Conditional Statement

- Conditional Statement: $p \rightarrow q$
- Converse: $q \rightarrow p$
- Example:
  - $p \rightarrow q$: If I am going to town, then it is raining.
  - $q \rightarrow p$: If it is raining, then I am going to town.
- Truth table

p	q	$\boldsymbol{p \rightarrow q}$	$\boldsymbol{q \rightarrow p}$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Inverse of Conditional Statement

- Conditional statement: $p \rightarrow q$
- Inverse: $\neg p \rightarrow \neg q$
- Example:
  - $p \rightarrow q$: If I am going to town, then it is raining.
  - $\neg p \rightarrow \neg q$: If I am not going to town, then it is not raining.
- Truth table

p	q	$\boldsymbol{p \rightarrow q}$	$\boldsymbol{\neg p \rightarrow \neg q}$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Contrapositive of Conditional Statement

- Conditional statement: $p \rightarrow q$
- Contrapositive: $\neg q \rightarrow \neg p$
- Example:
  - $p \rightarrow q$: If I am going to town, then it is raining.
  - $\neg q \rightarrow \neg p$: If it is not raining, then I am not going to town.
- Truth table

p	q	$\boldsymbol{p \rightarrow q}$	$\boldsymbol{\neg q \rightarrow \neg p}$
T	T	T	T
T	F	F	F
F	T	T	T
F	T	T	T

When two compound propositions always have the same truth value we call them *equivalent*.

Biconditional Operator

【Definition】 Let p and q be propositions. The biconditional statement $p \leftrightarrow q$, is the proposition "p if and only if (iff) q."

- Equivalent expressions:
  - ***p* is necessary and sufficient for *q***
  - **if *p* then *q* and conversely**
- Truth table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Both *p* and *q* must have the same truth value for $p \leftrightarrow q$ to be true.

Precedence of Logical Operators

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- Parentheses get the highest precedence
- Then $\neg \quad \wedge \quad \vee \quad \rightarrow \quad \leftrightarrow$

TABLE 8
Precedence of Logical Operators.

Operator	*Precedence*
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

- Examples:

  $p \wedge q \vee r$ means $(p \wedge q) \vee r$, not $p \wedge (q \vee r)$

  $p \vee q \rightarrow r$ means $(p \vee q) \rightarrow r$

Truth Table

- Truth Table: visual display of the relationship between the truth values of propositions
- Columns:
  - One column for each propositional variable
  - One column for each compound proposition
- Rows:
  - Each row contains a possible configuration of truth values of variables and the results of compound propositions
  - For n propositional variables there will be 2^n rows

Truth Table

$$(p \wedge q) \rightarrow r$$

Solution:

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	T
F	T	T	F	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

Logic and Bit Operations

- *Bit*: two possible values , 0 and 1
- A *Boolean variable* is one whose value is either true or false.
- A Boolean variable can be represented using a bit.
- *Bit operations* correspond to logical operations of Boolean variables.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
F	F	F	F	F
F	T	T	F	T
T	F	T	F	T
T	T	T	T	F

F→0 , T →1

Bitwise Operations

- A *bit string* is a sequence of zero or more bits.
- The *length* of this string is the number of bits in the string.
- *Bitwise operations* are bit operations extended to bit strings.
- Bitwise operations of two strings of the same length:
  - bitwise OR
  - bitwise AND
  - bitwise XOR

Bitwise Operations

Example: Find the bitwise OR, bitwise AND, and bitwise XOR of the bit strings 01 1011 0110 and 11 0001 1101.

01 1011 0110	
11 0001 1101	
11 1011 1111	bitwise OR
01 0001 0100	bitwise AND
10 1010 1011	birwise XOR

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Translating English sentences

- Example: How can the following sentence be translated into a logical expression?

  "*You can access the Internet from campus only if you are a computer science major or you are not a freshman.*"

Solution:

Let a: "you can access the Internet from campus,"
c: "you are a computer science major,"
f: "you are a freshman."

This sentence can be represented as

$$a \rightarrow (c \vee \neg f)$$

System Specifications

- Example: Express in logical expression

  "*The automated reply cannot be sent when the file system is full.*"

Solution:

Let p: "The automated reply can be sent"
q: "The file system is full"

This sentence can be represented as

$$q \rightarrow \neg p$$

Consistent System Specifications

【Definition】 A list of propositions is *consistent* if it is possible to assign truth values to the proposition variables so that each proposition is true.

- Example: Are these specifications consistent?
  - **"The diagnostic message is stored in the buffer or it is retransmitted."**
  - **"The diagnostic message is not stored in the buffer."**
  - **"If the diagnostic message is stored in the buffer, then it is retransmitted."**

Solution:

Let *p*: "The diagnostic message is stored in the buffer"
q: "The diagnostic message is retransmitted"

$$p \vee q,\ \neg p,\ p \rightarrow q$$

When *p* is false and *q* is true all three statements are true.
So these specifications are consistent.

Logic Puzzles

- Example: **An island has two kinds of inhabitants, *knights*, who always tell the truth, and *knaves*, who always lie. You go to the island and meet A and B.**
  - **A says "B is a knight."**
  - **B says "The two of us are of opposite types."**

What are the types of A and B?

Solution:

Let *p*: "A is a knight"
** *q*: "B is a knight"**

$$p \leftrightarrow q,\ q \leftrightarrow (p \wedge \neg q) \vee (\neg p \wedge q)$$

Homework

Due date: Mar. 7 (Tuesday)

Ver. 8

Sec. 1.1 2(a, b, d, e ,f), 10(e,h), 16(d,e), 20(c), 24, 30(b), 34(f), 50, 52

Sec. 1.2 4, 10, 22

Ver. 7

Sec. 1.1 2(a, b, d, e ,f), 8(e,h), 14(d,e), 18(c), 22, 28(b), 32(f), 46, 48

Sec. 1.2 4, 10, 18